

CALCELAR 25 mg Film Coated Tablet

Each film coated tablet contains:
Cinacalcet hydrochloride eq. to Cinacalcet 25 mg

✿ PRODUCT DESCRIPTION:

Light green, round, biconvex, film coated tablet, beveled on both sides with engraved CC 25 on one side and plain on the other

✿ MECHANISM OF ACTION:

Pharmacology

Cinacalcet exerts its effects by acting on calcium receptors on the surface of parathyroid cells. Calcium receptors control synthesis of PTH and proliferation of parathyroid cells, as well as secretion of PTH. The drug decreases the serum PTH level by acting on calcium receptors and mainly suppressing PTH secretion. Furthermore, in repeated-dose administration, its effect of suppressing proliferation of parathyroid cells is also considered to contribute to a decrease in serum PTH level.

Actions/Effects

a) Inhibition of PTH Secretion (*in vitro*)

Cinacalcet was shown to inhibit PTH secretion from bovine parathyroid cells and human parathyroid cells dose-dependently.

b) Suppression of proliferation of parathyroid cells

Following repeated-dose oral administration to partially nephrectomized rats, Cinacalcet was shown to inhibit proliferation of parathyroid cells and thereby suppress progression of parathyroid hyperplasia.

c) Decrease in the serum PTH and calcium levels

Following single-dose oral administration to normal rats and partially nephrectomized rats, Cinacalcet was shown to decrease the serum PTH and calcium levels dose-dependently.

d) Suppression of bone disorder

In patients with secondary hyperthyroidism, bone disorder develops due to increased serum PTH. Following repeated-dose oral administration to partially nephrectomized rats, Cinacalcet was shown to suppress symptoms associated with bone disorder due to increased serum PTH, such as marrow fibrosis, cortical osteoporosis, and decreases in cortical bone density and bone strength.

Pharmacokinetics

Absorption

For single oral administration of Cinacalcet at doses of 25, 50 and 100 mg during fasting condition, the plasma concentration of Cinacalcet increased dose dependently both on the non-dialysis and dialysis day for hemodialysis patients with secondary hyperthyroidism.

After multiple oral administration of Cinacalcet to hemodialysis patients with secondary hyperthyroidism, no tendency of increasing or decreasing over time on the plasma trough concentration and plasma concentration have reached a steady state following repeated-dose administration.

The effect of food on the pharmacokinetics of Cinacalcet is small.

Distribution

Protein-binding of Cinacalcet (25-100 ng/mL) in human plasma is 96.67 - 97.67% in males and 94.33 - 97.67% in females, respectively, and there is no sex difference.

Metabolism

Cinacalcet is extensively metabolized via *N*-dealkylation and oxidation of the naphthalene ring.

Elimination

Cinacalcet is excreted mainly in urine as metabolites.

Special populations

Renal and Hepatic impairment

Protein-binding is similar among normal, hepatic and renal impairment population which is 94.7 - 97.1% for normal and hepatic impairment population, and 92.7 - 95.1% for normal and renal impairment population, respectively. Cinacalcet is considered to be bound to albumin and shown to have a high affinity to the site II.

✿ INDICATION:

Indicated for secondary hyperparathyroidism patients undergoing maintenance dialysis

✿ DOSAGE AND ADMINISTRATION:

Oral

The starting dosage for adults is 25 mg of Cinacalcet once daily, to be orally administered.

With careful management of the patient's serum parathyroid hormone (PTH) and calcium levels, the dose may then be adjusted within a range of 25-75 mg once daily. If no improvement is found in PTH, the dose may be increase up to 100 mg once daily. If dose increase is required, dose should be increases by 25 mg at a time, at intervals of at least 3 weeks.

Precautions related to Dosage and Administration

1. Cinacalcet has an effect of decreasing calcium in blood. Therefore, it should be confirmed that the patient's serum calcium levels is not low [9.0 mg/dL (2.2 mmol/L) or more] prior to administration.
2. The serum calcium level should be determined once a week at the start of administration and during the dose adjustment period, and at least once every two weeks during the maintenance period. If serum calcium level decreases to 8.4 mg/dL (2.1 mmol/L) or less, the following measures should be taken.

Serum calcium level	Measures to be taken			
	Actions		Examinations	Dose increase/ Restart of treatment
	Administration of CALCELAR 25 mg Film Coated Tablet			
8.4 mg/dL (2.1 mmol/L) or less	In principle, the dose should not be increased (should be decreased as needed.)	Administration of Calcium or Vitamin D preparations should be considered.	The serum calcium should be determined at least once a week. It is recommended to perform electrocardiography.	If the dose is increased, the serum calcium should be confirmed to have recovered to 8.4 mg/dL (2.1 mmol/L) or more prior to dose increase.
7.5 mg/dL (1.9 mmol/L) or less	The drug should be withdrawn immediately.			If the treatment is restarted, the serum calcium level should be confirmed to have recovered to 8.4 mg/dL (2.1 mmol/L) or more prior to restart, and then the initial dose should be set at the pre-withdrawal level or less.

The serum level should be determined before administration of CALCELAR 25 mg Film Coated Tablet so that the effect and safety of the drug may be properly evaluated. Furthermore, it is recommended that corrected serum calcium values should be used as a guide in patients with hypoalbuminemia.

3. The serum PTH should be periodically determined so that it may be maintained at the target level for management. It is recommended that the serum PTH level should be determined twice a month at the start of administration and during the dose adjustment period (about 3 months after the start of administration), and at least once a month after the serum PTH level is confirmed to be almost stable. The serum PTH level should be determined before administration of CALCELAR 25 mg Film Coated Tablet so that the safety of the drug may be properly evaluated.

✿ WARNING AND PRECAUTION:

Careful administration in the following patients:

1. Patients with hypocalcemia [Hypocalcemia may be aggravated.] (See Important Precautions.)
2. Patients with seizure or a history of seizure
3. Patients with hepatic function disorder [The exposure amount will be increased since Cinacalcet is metabolized in the liver.]
4. Patients with gastrointestinal hemorrhage and/or gastrointestinal ulcer or a history of gastrointestinal hemorrhage and/or gastrointestinal ulcer. [The symptoms may worsen or recur.]

Important Precautions

1. During treatment with CALCELAR 25 mg Film Coated Tablet, sufficient caution should be exercised to avoid hypocalcemia by periodical measurement of the serum calcium level. If hypocalcemia occurs or may occur, it should be taken into account the use of Calcium or Vitamin D preparations, as well as to reduce the dose of CALCELAR 25 mg Film Coated Tablet (See Precautions related to Dosage and Administration.). If administration of calcium or vitamin D preparations is discontinued during the treatment with CALCELAR 25 mg Film Coated Tablet, caution should be exercised for possible occurrence of hypocalcemia. Symptoms likely to be associated with hypocalcemia such as prolonged QT interval, numbness, cramping, feeling unwell, arrhythmia, decreased blood pressure, etc. have been reported in clinical studies.
2. At the start of administration and during the dose adjustment period, the patient's symptoms should be frequently monitored and caution should be exercised for possible occurrence of adverse reactions.

Pediatric use

The safety of Cinacalcet in low birth weight infants, newborns, sucklings, infants and children has not been established.

Use in Elderly

Caution should be exercised when Cinacalcet is administered to patients aged 65 years or older, because they have been reported to show higher incidences of adverse reactions (prolonged QT interval in particular) than those younger than 65 years. If any adverse reactions are observed, appropriate measures such as reducing the dose should be taken.

Other Precautions

1. In an overseas clinical study in which Cinacalcet was used in patients with chronic renal failure accompanied by secondary hyperthyroidism who had not yet started dialysis, it has been reported that the blood calcium level tended to be lower than the lower limit of the normal range (8.4 mg/dL) compared to that in patients receiving dialysis. Use of the drug in patients not yet on dialysis has not been approved.
2. It has been reported abroad that adynamic bone disease occurred due to an excessive decrease in PTH following administration of Cinacalcet. It has been reported abroad that hungry bone syndrome accompanied by hypocalcaemia and hypophosphatemia occurred due to a rapid decrease in PTH following administration of Cinacalcet.

☼ PREGNANCY AND LACTATION:

Pregnancy

It is recommended not to use Cinacalcet in pregnant women or in women who may possibly be pregnant. The drug should be used in such patients only if the anticipated therapeutic benefits outweigh the potential risks associated with the treatment. The safety of Cinacalcet during pregnancy has not been established.

Hypocalcaemia, suppressed weight gain and decreased food consumption in mothers, as well as decreased weight in fetuses, were observed in animal studies (using rats and rabbits). Furthermore, Cinacalcet was reported to be transferred through the placenta in animal studies (using rats and rabbits).

Breastfeeding

Treatment with Cinacalcet should be avoided in nursing mothers. When the use is necessary in such women, breastfeeding should be discontinued.

In animal studies (using rats), Cinacalcet was reported to be transferred into milk. Furthermore, suppressed weight gain was observed in newborns on breastfeeding.

☼ SIDE EFFECT:

Clinically significant adverse reactions

1. **Hypocalcemia/ decreased serum calcium (14.7%):** Symptoms considered attributable to hypocalcemia (prolonged QT interval, numbness, cramping, feeling unwell, arrhythmia, decreased blood pressure, seizure, etc.) may occur. Therefore, the serum calcium level should be determined once a week at the start of administration and during the dose adjustment period, and at least once every two weeks during the maintenance period. If any abnormalities are observed, the serum calcium level should be confirmed and administration of calcium or vitamin D preparations taken into consideration. Moreover, it should also be considered to reduce the dose of CALCELAR 25 mg Film Coated Tablet or discontinue the treatment as needed (see Precautions related to Dosage and Administration).
2. **Prolonged QT interval (5.8%):** The QT interval may be prolonged. If any abnormalities are observed, the serum calcium level should be confirmed and administration of calcium or vitamin D preparations taken into consideration. Moreover, it should also be considered to reduce the dose of CALCELAR 25 mg Film Coated Tablet or discontinue the treatment as needed.
3. **Gastrointestinal hemorrhage, gastrointestinal ulcer (incidence unknown):** Gastrointestinal hemorrhage and/or gastrointestinal ulcer may develop. Patients must therefore be carefully monitored. If any abnormalities are observed, administration should be discontinued and appropriate measures taken.
4. **Decreased level consciousness (0.2%), temporary loss of consciousness (incidence unknown):** Decreased level of consciousness, temporary loss of consciousness, etc. may occur. If any abnormalities are observed, appropriate measures including discontinuation of the treatment should be taken.
5. **Sudden death (0.3%):** Unexplained sudden death has been reported in patients treated with Cinacalcet.

Other adverse reactions

If any of the following symptoms are observed, appropriate measure such as dosage reduction or discontinuation should be taken.

System Organ Class	Incidence (%)		
	≥ 5%	1% < 5%	< 1% or Incidence unknown*
Gastrointestinal	Nausea/vomiting, gastric discomfort, anorexia, abdominal distension	Upper abdominal pain, abdominal pain, abdominal discomfort, diarrhea, constipation, gastric ulcer, gastroduodenitis, reflux esophagitis, gastrointestinal disorder	Stomatitis, gastroenteritis, fecal occult blood, hemorrhoids, epigastric discomfort, hiatal hernia
Cardiovascular		Increased blood pressure, arrhythmia	Decreased blood pressure, myocardial infarction, myocardial ischemia, ventricular extrasystoles, supraventricular extrasystoles, atrial fibrillation, palpitations, tachycardia
Psycho-Neurological		Numbness, dizziness, headache, insomnia	
Musculoskeletal		Muscle spasm, pain in extremities	Arthralgia, myalgia, stiffness
Metabolic			Increased CK (CPK), increased LDH, increased blood sugar, hyperlipidemia, increased total cholesterol
Sensory		Dysgeusia	
Hepatic		Increased ALP	Abnormal hepatic function (Increased ASR (GOT)/ALT (GPT), increased bilirubin*, increased gamma-GTP*)
Ophthalmic			Conjunctival hemorrhage, dry eye
Skin		Pruritus	Rash, alopecia, subcutaneous hemorrhage
Endocrine			Goiter
Hematologic		Anemia	Decreased platelets
Others		Malaise, feeling unwell	Chest discomfort, chest pain, weakness, edema, shunt occlusion, thirst, pyrexia, erectile dysfunction

☼ EFFECTS ON ABILITY TO DRIVE AND USE MACHINE:

CALCELAR 25 mg Film Coated Tablet may have major influence on the ability to drive a vehicle or operate heavy machinery. Dizziness and seizures have been reported by patients taking CALCELAR 25 mg Film Coated Tablet.

☼ CONTRAINDICATION:

Patients with a history of hypersensitivity to any of the ingredients of CALCELAR 25 mg Film Coated Tablet.

☼ DRUG INTERACTION:

Cinacalcet should be administered with care when co-administered with the following drugs:

Drugs	Signs, Symptoms, Treatment	Mechanism, and Risk Factors
Azole antifungals <i>Itraconazole, etc.</i> Macrolide antibiotics <i>Erythromycin, Clarithromycin, etc.</i> Amiodarone hydrochloride Grapefruit juice	The plasma concentration of Cinacalcet may increase and the effect of the drug may enhance.	Cinacalcet is metabolized in part by CYP3A4. Therefore, the co-administration of CYP3A4 inhibitors listed in the left column, with Cinacalcet have a potency to inhibit the Cinacalcet metabolism and lead to an increase of plasma Cinacalcet concentration.

Tricyclic antidepressants <i>Amitriptyline hydrochloride,</i> <i>Imipramine hydrochloride, etc.</i> Butyrophenone Antipsychotics <i>Haloperidol, etc.</i> Flecainide Vinblastine	The blood concentration of these drugs may increase.	Cinacalcet is an inhibitor of CYP2D6. Therefore, the co-administration of Cinacalcet with CYP2D6 substrates listed in the left column have a potency to inhibit the metabolism of those CYP2D6 substrates and lead to an increase of their blood concentrations.
Calcitonin Biphosphate bone absorption suppressors <i>Pamidronate sodium,</i> <i>Alendronate sodium hydrate,</i> <i>Incadronate disodium, etc.</i> Corticosteroids <i>Cortisol,</i> <i>Prednisolone,</i> <i>Dexamethasone, etc.</i>	The serum calcium level may decrease.	The blood calcium-decreasing effect of Cinacalcet may be strengthened.
Digitoxin, Diazepam, etc.	The plasma concentration of Cinacalcet may be affected.	The degree of plasma protein-binding of Cinacalcet is high. (See Pharmacokinetics.)

⌘ **OVERDOSE AND TREATMENT:**

Symptoms

Overdosage of Cinacalcet is considered to cause hypocalcemia. In the case of overdosage, the patient should be monitored for any signs or symptoms of hypocalcemia.

Treatment

If hypocalcemia occurs or may occur, drip infusion of calcium preparations should be taken into consideration. As Cinacalcet is highly protein-bound, hemodialysis is not an effective treatment for overdosage.

⌘ **STORAGE:**

Store at temperature of not more than 30°C.

⌘ **DOSAGE FORM AND PACKAGING AVAILABLE:**

25 mg Tablet, Blister 10x10's

⌘ **DATE OF REVISION:**

December 18, 2023

Manufactured by:
UNISON LABORATORIES CO., LTD.
39 Moo 4, Klong Udomcholjorn, Muang Chachoengsao,
Chachoengsao 24000 Thailand